# Marco Zappoli

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Master's graduate in Aeronautical Engineering from Politecnico di Milano with a strong background in aerodynamics and turbulence. Experienced in structured problem-solving and collaborative work through academic projects, with leadership and discipline developed through competitive tennis at Italian second division level.

## Education

**Master of Science in Aeronautical Engineering - Politecnico di Milano (Milan, Italy)** **Sep 2023 - Mar 2026**
- Final mark: 110/110 cum laude
- GPA: 29.47/30
- Thesis: Clustering and preferential sampling in homogeneous shear turbulence
  (Advisor: Dr. Alessandro Chiarini, Co-advisor: Prof. Maurizio Quadrio)
- Relevant courses: Aerodynamics, Turbulence: Physics and Modeling, Numerical Modeling of Differential Problems, Computational Fluid Dynamics, Compressible Fluid Dynamics, Structural Dynamics and Aeroelasticity

**Bachelor of Science in Aerospace Engineering - Politecnico di Milano (Milan, Italy)** **Sep 2020 - Jul 2023**
- Final mark: 105/110
- GPA: 26.77/30

**High School Diploma - Liceo Scientifico B. Cavalieri (Verbania, Italy)** **Sep 2015 - Jun 2020**
- Final mark: 95/100

## Publication

Manuscript in preparation for submission to Journal of Fluid Mechanics.

## Skills

**CAD & CFD Tools:** SolidWorks, Solid Edge, SU2, gmsh
**Programming:** MATLAB, CPL, Python, Fortran
**Tools:** LaTeX, Microsoft Office
**Operating Systems:** Windows, Linux (Ubuntu), High-Performance Computing (CINECA cluster usage)
**Languages:** English (fluent), Italian (native)

## Working Experience

**Food Sales Associate – Villiborghi SAS di Villiborghi Daniele & C.** **Jun 2019 - Aug 2019**
- Food sales associate and primary customer contact in an international market (Italian and German clientele)

## Relevant Projects

• **FOSSER: Formation-flight Optimization for a SubSonic mission with Extended Range** **A.Y. 2024/2025**
Analyzed fighter aircraft formation aerodynamics using AVL, optimizing wing design and positioning to reduce drag and extend range, with validation against experimental data and consideration of aeroelastic trade-offs.

• **Experimental Wind Tunnel Testing: Advanced Measurement and Flow Characterization** **A.Y. 2024/2025**
Performed wind tunnel testing on airfoils and eVTOL propellers, including force measurements, pressure and wake analysis, and PIV-based flow field reconstruction using MATLAB.

• **Impact of Ice Formation on the Aerodynamic Performance of the NACA 23012 Airfoil** **A.Y. 2024/2025**
Performed RANS CFD simulations to assess ice accretion effects on airfoil aerodynamics, including flow separation and stall. Developed automated meshing workflows and conducted turbulence/transition model sensitivity analyses.

• Additional projects, reports, PDFs, and thesis work are available through the personal website linked in the header.

## Additional Information

**Certification:** IELTS Academic – Overall 8.0 (L: 8.5, R: 9.0, W: 7.0, S: 7.0); Test date: July 2026
**Membership:** AVIS blood donor - Verbania (2022-present)
**Interests:** Tennis, Formula 1, Fitness, Reading, Cinema, Aviation and Aerospace

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